Full Reading Guide

Agentic AI for Utility Operators — integrated power-system engineering tutorial. Evidence reviewed 17 October 2026. The full reading guide is separate from the timed main talk. Corporate issuers are listed when no individual author is recorded. Original figures, page locations and image-source credits are included. Attribution alone is not a blanket reuse license.

[B01] IEEE Communications Society: official resources and network presentation template

Authors / issuer: IEEE Communications Society

Publisher / venue: IEEE Communications Society

Date: Reviewed 17 October 2026

Theme assets: comsoc-lockup.png and image7.png; independent educational adaptation

Full source URL: https://www.comsoc.org/about/brand-experience/resources

[DEF03] Building effective agents

Authors / issuer: Erik S., Barry Zhang

Publisher / venue: Anthropic Engineering

Date: 2024-12-19

Type: Official practitioner architecture account by the system maintainers

Location: What are agents?; Agents; Workflow: Routing; Workflow: Orchestrator-workers; Workflow: Evaluator-optimizer

Version note: The live page now warns that tooling has changed since the original publication. Use its conceptual distinction, not its present SDK list as a claim about the 2024 implementation.

Full source URL: https://www.anthropic.com/engineering/building-effective-agents

[DEF02] Intelligent agents: theory and practice

Authors / issuer: Michael Wooldridge, Nicholas R. Jennings

Publisher / venue: The Knowledge Engineering Review, 10(2), pp. 115–152

Date: 1995

Type: Primary theoretical/architectural research review by the authors

Location: Section 1.1, What is an Agent?; Section 1.1.1, A Weak Notion of Agency

Version note: The author-hosted HTML is an earlier draft stamped November 1994. Use the final 1995 PDF for the bibliographic citation; the HTML makes the definition sections searchable. The final PDF is image-based in the web extractor, so section locators are safer than an unverified PDF page number.

Full source URL: https://doi.org/10.1017/S0269888900008122

Full paper PDF: https://www.cs.ox.ac.uk/people/michael.wooldridge/pubs/ker95.pdf

[NV01] Building for the Rising Complexity of Agentic Systems with Extreme Co-Design

Authors / issuer: Eduardo Alvarez, Benjamin Klieger, Graham Steele

Publisher / venue: NVIDIA Technical Blog

Date: May 5, 2026

Type: Official vendor architectural explanation; conceptual figure

Location: Transition to agents from chatbots, Figure 1

Full source URL: https://developer.nvidia.com/blog/building-for-the-rising-complexity-of-agentic-systems-with-extreme-co-design/

[GEN01] Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile

Authors / issuer: National Institute of Standards and Technology

Publisher / venue: NIST AI 600-1

Date: July 2024

Type: Official primary definition and risk-management profile

Location: Introduction, printed p.1, footnote 1; retrieved 18 October 2026

Full source URL: https://doi.org/10.6028/NIST.AI.600-1

Full paper PDF: https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf

[DEF01] Artificial Intelligence: A Modern Approach, Chapter 2: Intelligent Agents

Authors / issuer: Stuart J. Russell, Peter Norvig

Publisher / venue: Prentice Hall, second edition, Chapter 2, pp. 32–58

Date: 2003

Type: Canonical authored textbook; primary exposition of its agent framework

Location: Section 2.1, printed pp. 32-34 / PDF pp. 1-3; Figure 2.1, printed p. 33 / PDF p. 2; Section 2.2, printed pp. 35-37 / PDF pp. 4-6

Version note: Despite its current /4th-ed/ URL, the verified sample chapter is the older chapter with printed pages 32-58 and 2002 directory timestamp. Do not cite this PDF as fourth-edition text. The official second-edition page identifies the edition and Prentice Hall; the final book citation is 2003.

Full source URL: https://aima.cs.berkeley.edu/2nd-ed/

Full paper PDF: https://aima.cs.berkeley.edu/4th-ed/pdfs/newchap02.pdf

[DEF06] Tools

Authors / issuer: LangChain maintainers

Publisher / venue: Docs by LangChain, Python open-source documentation

Date: Accessed 17 October 2026

Type: Official tool-interface documentation

Full source URL: https://docs.langchain.com/oss/python/langchain/tools

[NV02] Introduction to LLM Agents

Authors / issuer: Tanay Varshney

Publisher / venue: NVIDIA Technical Blog

Date: November 30, 2023

Type: Official vendor architectural explanation; conceptual figure

Location: What is an AI agent? Figure 1: General components of an agent

Full source URL: https://developer.nvidia.com/blog/introduction-to-llm-agents/

[DEF04] LangGraph overview

Authors / issuer: LangChain maintainers

Publisher / venue: Docs by LangChain, Python open-source documentation

Date: Undated live documentation; accessed 2026-10-17

Type: Official framework and tool-interface documentation; two pages in one documentation collection

Location: LangGraph overview: opening paragraphs and product-role list; Tools: opening definition and Basic tool definition

Full source URL: https://docs.langchain.com/oss/python/langgraph/overview

[LG04] Workflows and agents

Authors / issuer: LangChain documentation maintainers

Publisher / venue: Docs by LangChain, Python LangGraph documentation

Date: Undated live documentation; accessed October 18, 2026

Type: Official framework concepts and source-authored illustrations

Location: Opening Agent Workflow illustration (selected agent panel); LLM augmentations; workflow patterns

Full source URL: https://docs.langchain.com/oss/python/langgraph/workflows-agents

[LG03] LangGraph Interrupts

Authors / issuer: LangChain documentation maintainers

Publisher / venue: Official LangChain documentation

Date: Accessed 2026-10-17; live documentation

Evidence type: Official Python runtime documentation

Full source URL: https://docs.langchain.com/oss/python/langgraph/interrupts

[PA02] PowerAgent — Open-Source Agentic AI for Power Grids

Authors / issuer: Power and AI Initiative / PowerAgent community, Harvard SEAS

Publisher / venue: Official Harvard project website

Date: Live project page reviewed 17 October 2026

Evidence type: Primary project scope, maintenance and demonstration descriptions

Full source URL: https://poweragent.seas.harvard.edu/

[PD01] PowerDAG: Supervisory Agentic AI System for Automating Distribution Grid Analysis

Authors / issuer: Emmanuel O. Badmus, Amritanshu Pandey

Publisher / venue: arXiv preprint, version 4

Date: 5 July 2026 (v4); first submitted 18 March 2026

Evidence type: Full primary research paper; v4 inspected

Full source URL: https://arxiv.org/pdf/2603.17418v4

[N01] Field Notes / HyperQ deployment in PJM's Cycle 1

Authors / issuer: Tapestry

Publisher / venue: Tapestry

Date: 2026-06-11; deployment entry 2026-04-29

Type: supplier primary deployment/results report

Full source URL: https://www.tapestryenergy.com/en/projects/field-notes-hyperq-deployment-in-pjm-s-cycle-1

[N02] How agentic AI can help grid operators speed up interconnection

Authors / issuer: Tapestry

Publisher / venue: Tapestry

Date: 2025-12-08 (linked chronological entry in N01); live article reviewed 2026-10-17

Type: supplier primary technical white paper

Full source URL: https://www.tapestryenergy.com/en/projects/how-agentic-ai-can-help-grid-operators-speed-up-interconnection

[N06] Successful Interconnection Reforms, Other PJM Initiatives Seek To Maximize Electricity Supplies

Authors / issuer: PJM Interconnection

Publisher / venue: PJM Interconnection

Date: 2026-03-16

Type: grid-operator primary progress report

Full source URL: https://insidelines.pjm.com/connected/

[C01] Exploring AI to support smarter grid management

Authors / issuer: California ISO

Publisher / venue: California ISO

Date: 2025-07-28

Type: customer primary source

Full source URL: https://www.caiso.com/about/news/energy-matters-blog/exploring-ai-to-support-smarter-grid-management

[C02] Generative and Agentic AI in Grid Operations: Field Lessons from OATI, CAISO, PNNL, and EPRI

Authors / issuer: OATI

Publisher / venue: OATI

Date: 2026 (exact publication day not displayed)

Type: vendor primary field report

Full source URL: https://www.oati.com/ai-for-energy/

[I06] California ISO control room, image published by OATI: California ISO control room with operators seated at curved desks facing a long wall of grid displays

Authors / issuer: OATI (California ISO control-room photograph)

Publisher / venue: OATI (California ISO control-room photograph)

Date: Image accessed 17 October 2026

Full source URL: https://www.oati.com/ai-genie/

[U02] AWS Launches Agentic Grid Planning Program to Accelerate Interconnection Studies

Authors / issuer: Amazon Web Services

Publisher / venue: Amazon Web Services

Date: 2026-09-17

Type: vendor primary announcement with named Duke Energy collaboration and testimony

Full source URL: https://press.aboutamazon.com/aws/2026/9/aws-launches-agentic-grid-planning-program-to-accelerate-interconnection-studies

[U03] Agentic Grid Planning on AWS

Authors / issuer: Amazon Web Services

Publisher / venue: Amazon Web Services

Date: Undated; reviewed 17 October 2026

Type: vendor primary program and workflow description

Full source URL: https://aws.amazon.com/energy-utilities/agentic-grid-planning/

[I03] Duke Energy, illumination, August 2, 2022: Duke Energy grid operator Howard Greene seated before many control-room displays

Authors / issuer: Duke Energy / illumination

Publisher / venue: Duke Energy / illumination

Date: Image accessed 17 October 2026

Full source URL: https://illumination.duke-energy.com/articles/meet-the-team-that-monitors-the-grid-and-keeps-the-lights-on

[N03] Siemens' Gridscale X redefines system operations and agentic transmission planning

Authors / issuer: Siemens

Publisher / venue: Siemens

Date: 2026-05-05

Type: vendor primary product announcement

Full source URL: https://press.siemens.com/global/en/pressrelease/siemens-gridscale-x-redefines-system-operations-and-agentic-transmission-planning

[N04] National Grid Full Year Results 2025/26: Video Webcast Transcript

Authors / issuer: National Grid

Publisher / venue: National Grid

Date: 2026-05-14

Type: utility primary investor-results transcript

Full source URL: https://www.nationalgrid.com/document/577036/download

[N09] On demand: Trust-driven medallion architecture: How to make your CDEs ready for AI

Authors / issuer: Session speakers: Saptaparna Gupta (NYPA), Oleksandra Bovkun (Databricks), Eliza Pochinova (Ataccama)

Publisher / venue: Ataccama

Date: 2026-10-14 (on-demand page); event page currently lists 2026-10-13

Type: Original vendor-hosted technical session report with named utility participant

Full source URL: https://www.ataccama.com/webinar/on-demand-trust-driven-medallion-architecture

[N10] Trust-driven medallion architecture: how to make your CDEs ready for AI

Authors / issuer: Saptaparna Gupta (NYPA), Oleksandra Bovkun (Databricks), Eliza Pochinova (Ataccama)

Publisher / venue: Ataccama

Date: 2026-10-13, 11:00 AM EDT (date displayed on fetched event page)

Type: Original technical event listing with named utility speaker

Full source URL: https://www.ataccama.com/events/trust-driven-medallion-architecture

[N07] Distribution System Modeling and Automation Engineer (Hybrid Schedule)

Authors / issuer: Eversource Energy (employer)

Publisher / venue: Eversource Energy

Date: 2026-10-05

Type: Utility primary engineering recruitment disclosure; capability-building evidence

Full source URL: https://jobs.eversource.com/job/east-berlin/distribution-system-modeling-and-automation-engineer-hybrid-schedule/7774/101576695504

[N11] Eversource Energy Collaborates with MathWorks to Incorporate Renewable Energy Sources into System Planning Process Using Probabilistic Load Flow Automation

Authors / issuer: MathWorks; named Eversource engineer John Kreso quoted

Publisher / venue: MathWorks

Date: 2025-04-29

Type: Supplier primary engineering deployment report with named utility testimony; conventional automation

Full source URL: https://www.mathworks.com/company/newsroom/eversource-energy-collaborates-with-mathworks-to-incorporate-renewable-energy-sources-into-system-planning-process-using-probabilistic-load-flow-automation.html

[C07] GE Vernova Launches GridOS for Distribution, the Industry's First Unified Solution for Grid Orchestration

Authors / issuer: GE Vernova

Publisher / venue: GE Vernova

Date: 2026-02-03

Type: vendor primary announcement with named utility statement

Full source URL: https://www.gevernova.com/news/press-releases/ge-vernova-launches-gridos-distribution-industrys

[I01] GE Vernova: GE Vernova GridOS promotional control-room composite with people and grid software displays

Authors / issuer: GE Vernova

Publisher / venue: GE Vernova

Date: Image accessed 17 October 2026

Full source URL: https://www.gevernova.com/software/products/gridos

[Q01] An Analysis of Grid Operators

Authors / issuer: Corey Fallon, Brett Jefferson, Tyler Ryan

Publisher / venue: Pacific Northwest National Laboratory

Date: 2023-04-19

Type: Primary research presentation; 20-operator survey

Location: Methods PDF p.5; workload result PDF p.9. Exact survey question and task denominator are not supplied.

Full source URL: https://www.pnnl.gov/sites/default/files/media/file/PNNL_workshop_Presentation_Human_Factors_Virtual_Workshop_March_2023.pdf

[O01] What Happens in the Control Room

Authors / issuer: PJM Interconnection

Publisher / venue: PJM Interconnection

Date: 2026-04-16

Type: Official operator fact sheet

Full source URL: https://www.pjm.com/-/media/DotCom/about-pjm/newsroom/fact-sheets/what-happens-in-the-control-room-fact-sheet.pdf

[I02] Duke Energy, illumination, August 2, 2022: Duke Energy employees in an Energy Control Center, with one pointing toward a display

Authors / issuer: Duke Energy / illumination

Publisher / venue: Duke Energy / illumination

Date: Image accessed 17 October 2026

Full source URL: https://illumination.duke-energy.com/articles/meet-the-team-that-monitors-the-grid-and-keeps-the-lights-on

[O05] Control center modernization is enhancing reliability

Authors / issuer: John Phipps

Publisher / venue: California ISO

Date: 2025-09-10

Type: Official operator article and tool screenshot

Full source URL: https://www.caiso.com/about/news/energy-matters-blog/control-center-modernization-is-enhancing-reliability

[I09] GridOS Visual Intelligence product screens

Authors / issuer: GE Vernova

Publisher / venue: GE Vernova

Date: Accessed 17 October 2026

Full source URL: https://www.gevernova.com/software/products/gridos/visual-intelligence

[LG01] LangGraph Graph API overview

Authors / issuer: LangChain documentation maintainers

Publisher / venue: Official LangChain documentation

Date: Accessed 2026-10-17; live documentation

Evidence type: Official Python runtime documentation

Full source URL: https://docs.langchain.com/oss/python/langgraph/graph-api

[E01] X-GridAgent: An LLM-Powered Agentic AI System for Assisting Power Grid Analysis

Authors / issuer: Yihan (Logon) Wen, Xin Chen

Publisher / venue: arXiv preprint

Date: 2025-12-23 (v1)

Evidence type: Original research; full v1 PDF

Full source URL: https://arxiv.org/pdf/2512.20789v1

[R01] Integrating Agentic Artificial Intelligence with High-Performance Computing for Grid Planning

Authors / issuer: Samim Konjicija, Slaven Peles

Publisher / venue: arXiv; EnergyCon 2026 proceedings forthcoming as described by authors

Date: 3 September 2026; v1

Type: Research preprint; not a utility deployment

Location: Abstract and framework description: LLM-directed analyses through ExaGO; University of Sarajevo and Oak Ridge National Laboratory affiliations

Full source URL: https://arxiv.org/abs/2609.04544v1

Full paper PDF: https://arxiv.org/pdf/2609.04544v1

[L01] GridGuard current project: README and graph source snapshot

Authors / issuer: GridGuard project / local source snapshot

Publisher / venue: GridGuard project / local source snapshot

Date: Snapshot 17 October 2026; see research/GridGuard_source_snapshot.txt

Full source URL: research/GridGuard_source_snapshot.txt

[D01] DrAgent: An Agentic Approach to Fault Analysis in Power Grids Using Large Language Models

Authors / issuer: Barun Kumar Saha, Aarthi V, O. D. Naidu

Publisher / venue: 2025 International Conference on Artificial Intelligence in Information and Communication (ICAIIC), pp. 938–945

Date: 2025-02-18–21

Type: Peer-reviewed original power-system research; full conference-hosted PDF

Full source URL: https://www.manuscriptlink.com/society/kics/media?key=kics%2Fconference%2Ficaiic2025%2Fpresentation%2F1571098105.pdf

Publication: https://doi.org/10.1109/ICAIIC64266.2025.10920654

[E02] Enhancing LLMs for Power System Simulations: A Feedback-Driven Multi-Agent Framework

Authors / issuer: Mengshuo Jia, Zeyu Cui, Gabriela Hug

Publisher / venue: IEEE Transactions on Smart Grid, volume 16, issue 6, pages 5556–5572

Date: Published online 2025-07-18; issue November 2025; inspected full arXiv v3 dated 2025-05-19

Evidence type: Published original research; publisher metadata and full author preprint inspected

Full source URL: https://doi.org/10.1109/TSG.2025.3589114

Full paper PDF: https://arxiv.org/pdf/2411.16707v3

[E03] Grid-Agent: An LLM-Powered Multi-Agent System for Power Grid Control

Authors / issuer: Yan Zhang, Ahmad Mohammad Saber, Amr Youssef, Deepa Kundur

Publisher / venue: arXiv preprint

Date: 2025-09-08 (v3); first submitted 2025-08-07

Evidence type: Original research; full v3 PDF

Full source URL: https://arxiv.org/pdf/2508.05702v3

[P02] PowerBench: A Benchmark for Agentic Retrieval and Reasoning in Power Systems

Authors / issuer: Xijing Wang, Yinsheng Yao, Jinru Ding, Yidong Jiang, Ziwen Xu, Yiwen Jiang, Jie Xu, Dawei Cheng

Publisher / venue: arXiv

Date: 2026-09-28 (v1)

Type: Primary research preprint; synthetic retrieval benchmark

Location: Tables 2–4, PDF pp.5,7–8; Appendix B, D and F define tasks, metrics and limitations.

Full source URL: https://arxiv.org/abs/2609.34492v1

Full paper PDF: https://arxiv.org/pdf/2609.34492v1

[I04] Image published by NVIDIA with its EPRI Open Power AI Consortium announcement, March 20, 2025: Transmission towers and power lines with city lights in the background at dusk

Authors / issuer: NVIDIA

Publisher / venue: NVIDIA

Date: Image accessed 17 October 2026

Full source URL: https://blogs.nvidia.com/blog/open-power-ai-consortium/

[PA01] PowerAgent: A Road Map Toward Agentic Intelligence in Power Systems: Foundation Model, Model Context Protocol, and Workflow

Authors / issuer: Qian Zhang, Le Xie

Publisher / venue: IEEE Power and Energy Magazine, volume 23, issue 5, pages 93–101

Date: September–October 2025 issue; TechRxiv preprint DOI version 1, June 2025

Evidence type: Primary roadmap article; identity verified through Harvard author pages and indexed author preprint; full article download unavailable during this review

Full source URL: https://doi.org/10.1109/MPE.2025.3579718

[PA03] A Roadmap Towards Agentic Intelligence in Power Systems

Authors / issuer: Le Xie; slides credit Le Xie and Qian Zhang

Publisher / venue: ERCOT Technology and Security Committee guest presentation

Date: 8 December 2025

Evidence type: Full original 19-page author presentation inspected; roadmap and workflow demonstrations

Full source URL: https://www.ercot.com/files/docs/2025/12/01/4-Guest-Presentation-A-Roadmap-toward-Agentic-Intelligence-in-Power-Systems.pdf

[P01] PowerAgentBench-Dyn: A Benchmark for Agentic AI in Power System Dynamic Studies

Authors / issuer: Qian Zhang, Andrea Pomarico, Costas Mylonas, Magda Foti, Alberto Berizzi, Le Xie

Publisher / venue: arXiv

Date: 2026-06-18 (v1)

Type: Primary research preprint; prototype benchmark

Location: DMQ Tables I–II, PDF p.3; DSR Tables III–VI, pp.4–5.

Full source URL: https://arxiv.org/abs/2606.20401v1

Full paper PDF: https://arxiv.org/pdf/2606.20401v1

[PA06] PowerWF Grid Impact Evaluation — LangGraph workflow source

Authors / issuer: PowerAgent / PowerWF community contributors

Publisher / venue: Author-maintained source code, Grid Impact Evaluation/src/agent/graph.py

Date: Live source inspected 17 October 2026

Evidence type: Primary implementation: StateGraph, ToolNode, PowerWorld tool groups, conditional routing; inspected, not executed

Full source URL: https://github.com/Power-Agent/PowerWF/blob/main/Grid%20Impact%20Evaluation/src/agent/graph.py

[PA05] Power-Agent/PowerWF

Authors / issuer: PowerAgent community contributors

Publisher / venue: Author-maintained GitHub repository

Date: Live source repository reviewed 17 October 2026

Evidence type: Primary workflow source repository

Full source URL: https://github.com/Power-Agent/PowerWF

[P03] GridMind: An Agentic Workflow for Weather-Driven Power Grid Risk Assessment

Authors / issuer: D. Adrian Maldonado, Jonghwan Kwon, Sunash B. Sharma, Hongwei Jin, Stefano Fenu, Emil Constantinescu, David Sehloff, Parfait Gasana, Duane Rudolph Verner, Kibaek Kim

Publisher / venue: KDD 2026 Workshop on Scientific and Societal Agents / OpenReview

Date: August 2026 workshop; reviewed 17 October 2026

Type: Primary workshop paper; simulated workflow evaluation

Location: Architecture section 3, PDF p.2; seven-query trace and ablation Table1, PDF p.4.

Full source URL: https://openreview.net/forum?id=Cs5TBzzNym

Full paper PDF: https://openreview.net/pdf/5ad1c2afd7298f9e6ce3bc25f18e7165555e1ffd.pdf

[LG02] LangGraph Persistence

Authors / issuer: LangChain documentation maintainers

Publisher / venue: Official LangChain documentation

Date: Accessed 2026-10-17; live documentation

Evidence type: Official Python runtime documentation

Full source URL: https://docs.langchain.com/oss/python/langgraph/persistence

[LG05] Checkpointers

Authors / issuer: LangChain documentation maintainers

Publisher / venue: Docs by LangChain, Python LangGraph documentation

Date: Undated live documentation; accessed October 18, 2026

Type: Official framework persistence documentation and illustrations

Location: Opening Checkpoints illustration; Get state history / State illustration

Full source URL: https://docs.langchain.com/oss/python/langgraph/checkpointers

[PD02] DistGrid-AgentBench: A Distribution Grid Benchmark for Agentic AI

Authors / issuer: Emmanuel O. Badmus and contributors

Publisher / venue: Author-maintained GitHub repository

Date: Live repository reviewed 17 October 2026

Evidence type: Primary benchmark tasks, tool catalog and evaluator; README explicitly documents proprietary omissions

Full source URL: https://github.com/emmanuelbadmus/DistGrid-AgentBench

[R02] Agentic Artificial Intelligence for Power Systems: Strategies to Identify and Close Capability Gaps

Authors / issuer: Eve Tsybina, Samim Konjicija, Slaven Peles

Publisher / venue: arXiv

Date: 30 July 2026; v1

Type: Research preprint; results apply to the tested prototype and suite

Location: Six task-complexity levels and four grid scales; only the two lowest levels solvable on some tested sizes

Full source URL: https://arxiv.org/abs/2607.28822v1

Full paper PDF: https://arxiv.org/pdf/2607.28822v1

[VID01] LangSmith Studio: The first agent IDE

Authors / issuer: LangChain

Publisher / venue: LangChain

Date: Launch walkthrough linked 1 August 2024

Evidence type: Official platform video; not evidence of a named utility deployment

Full source URL: https://www.youtube.com/watch?v=pLPJoFvq4_M

Publication: https://www.langchain.com/blog/langgraph-studio-the-first-agent-ide

[VID02] Gridscale X PSS E: agentic transmission planning — product video

Authors / issuer: Siemens

Publisher / venue: Siemens

Date: Undated video; accessed 26 October 2026

Evidence type: Official platform video; not evidence of a named utility deployment

Full source URL: https://video.sw.siemens.com/embed/682ca322-5d0b-47a0-8d18-77bf9d0fb016

Publication: https://www.siemens.com/en-gb/products/pss-software/gridscale-x-pss-e/agentic-transmission-planning/

[P04] GridMind: LLMs-Powered Agents for Power System Analysis and Operations

Authors / issuer: Hongwei Jin, Kibaek Kim, Jonghwan Kwon

Publisher / venue: arXiv; published as SC Workshops 2025, DOI 10.1145/3731599.3767409

Date: 2025-09-02 (arXiv v1); SC Workshops 2025

Type: Primary research paper; early prototype

Location: section 4, Fig3 and Table1, PDF pp.4–5.

Full source URL: https://arxiv.org/abs/2509.02494v1

Full paper PDF: https://arxiv.org/pdf/2509.02494v1

[P03A] GridAI — AI Foundation Models & Agents for the Power Grid

Authors / issuer: Argonne GridAI project team

Publisher / venue: Argonne GridAI project team

Date: Live project page reviewed 17 October 2026

Type: Author-maintained project and publication index

Full source URL: https://argonne-gridfm.github.io/

[P01C] Power-Agent/PowerAgentBench

Authors / issuer: PowerAgent Community

Publisher / venue: PowerAgent Community

Date: Live author repository reviewed 17 October 2026

Type: Author-maintained benchmark code, cases and specifications

Location: Dynamic level1 harness and README; licensed PSS/E and DMView prerequisites.

Full source URL: https://github.com/Power-Agent/PowerAgentBench

[E04] VeraGrid-Agent: Tool-Augmented LLMs for Distribution Optimal Power Flow at the Grid Edge

Authors / issuer: Shivanshu Tripathi, Hamed Mohsenian-Rad, Maziar Raissi

Publisher / venue: arXiv preprint

Date: 2026-07-28 (v1)

Evidence type: Original research; full v1 PDF

Full source URL: https://arxiv.org/pdf/2607.25155v1

[E04C] Shivanshu-11/veragrid-llm-benchmark

Authors / issuer: Shivanshu Tripathi / VeraGrid-Agent authors

Publisher / venue: Author GitHub repository

Date: Reviewed 2026-10-17

Evidence type: Author implementation, benchmark and recorded-run repository

Full source URL: https://github.com/Shivanshu-11/veragrid-llm-benchmark

[R03] 7.2.4.1. Confidence intervals

Authors / issuer: National Institute of Standards and Technology, SEMATECH

Publisher / venue: NIST/SEMATECH e-Handbook of Statistical Methods

Date: Undated live handbook section; reviewed 17 October 2026

Type: Official statistical methods reference

Location: Wilson interval formula and exact binomial intervals. Tutorial intervals are calculated from R02 counts, not reported by the grid paper.

Full source URL: https://www.itl.nist.gov/div898/handbook/prc/section2/prc241.htm

[R04] EXACT BINOMIAL

Authors / issuer: National Institute of Standards and Technology

Publisher / venue: NIST Dataplot Reference Manual

Date: Last updated 16 November 2015; reviewed 17 October 2026

Type: Official statistical methods reference

Location: One-sided and two-sided exact binomial confidence limits. The tutorial derives the zero-failure bound from the binomial probability, with independent fixed-setting trials as an explicit assumption.

Full source URL: https://www.itl.nist.gov/div898/software/dataplot/refman2/auxillar/exacbino.htm

[T08] MCP Architecture overview

Authors / issuer: Model Context Protocol

Publisher / venue: Model Context Protocol

Date: 2026-07-28 documentation revision

Evidence type: Official protocol documentation

Full source URL: https://modelcontextprotocol.io/docs/2026-07-28/learn/architecture

[T09] MCP Security Best Practices

Authors / issuer: Model Context Protocol

Publisher / venue: Model Context Protocol

Date: 2026-07-28 documentation revision

Evidence type: Official protocol documentation

Full source URL: https://modelcontextprotocol.io/docs/2026-07-28/tutorials/security/security_best_practices

[U01] AES: Scaling energy safety audits with Claude on Google Cloud

Authors / issuer: Google Cloud

Publisher / venue: Google Cloud

Date: Undated; reviewed 17 October 2026

Type: vendor primary customer case study with named customer testimony

Full source URL: https://cloud.google.com/customers/aes

[U04] TC Energy increases office productivity with automation on Amazon Bedrock

Authors / issuer: Amazon Web Services

Publisher / venue: Amazon Web Services

Date: Undated; reviewed 17 October 2026

Type: vendor primary customer case study with named customer testimony

Full source URL: https://aws.amazon.com/solutions/case-studies/tcenergy-bedrock/

[O03] ISO now has expanded training and emergency operations facilities

Authors / issuer: Tricia Johnstone, Director, Operational Readiness

Publisher / venue: California ISO

Date: 2025-02-20

Type: Official operator organization article

Full source URL: https://www.caiso.com/about/news/energy-matters-blog/iso-now-has-expanded-training-and-emergency-operations-facilities

[C15] CESER Releases New Testbed to Advance LLM and Agentic AI Evaluation for Critical Infrastructure

Authors / issuer: U.S. Department of Energy

Publisher / venue: U.S. Department of Energy

Date: 2026-07-16

Type: government primary research-tool announcement

Full source URL: https://www.energy.gov/ceser/articles/ceser-releases-new-testbed-advance-llm-and-agentic-ai-evaluation-critical

[C11] EPRI Launches Effort to Reduce Risk in AI Power Sector Deployment

Authors / issuer: EPRI via GlobeNewswire

Publisher / venue: EPRI via GlobeNewswire

Date: 2026-08-06

Type: issuer-authored primary press release

Full source URL: https://www.globenewswire.com/news-release/2026/08/06/3340219/0/en/epri-launches-effort-to-reduce-risk-in-ai-power-sector-deployment.html

[C12] SAFERai.power

Authors / issuer: EPRI

Publisher / venue: EPRI

Date: Undated; reviewed 17 October 2026

Type: primary initiative and roadmap

Full source URL: https://msites.epri.com/saferai

[I05] Oak Ridge National Laboratory, U.S. Department of Energy: Researchers at ORNL GOAL facing a large display of U.S. grid and geographic data

Authors / issuer: Oak Ridge National Laboratory, U.S. Department of Energy

Publisher / venue: Oak Ridge National Laboratory, U.S. Department of Energy

Date: Image accessed 17 October 2026

Full source URL: https://www.ornl.gov/grid-operations-and-analytics-laboratory-goal/

[L02] Existing three-use-case GridGuard workshop: instructor route and synthetic-feeder context

Authors / issuer: Local workshop source snapshot

Publisher / venue: Local workshop source snapshot

Date: Snapshot 17 October 2026

Location: Voltage support, service restoration and cyber response are separate worked modules.

Full source URL: research/Follow_on_workshop_source.txt

[P02C] open-compass/PowerBench

Authors / issuer: OpenCompass / PowerBench authors

Publisher / venue: OpenCompass / PowerBench authors

Date: Live author-linked repository reviewed 17 October 2026

Type: Linked project repository; only README and LICENSE visible at review

Full source URL: https://github.com/open-compass/PowerBench

[P03M] KDD 2026 SciSoc LLM Workshop — Accepted Papers

Authors / issuer: KDD26SciSocLLM workshop organizers

Publisher / venue: KDD26SciSocLLM workshop organizers

Date: 2026 workshop listing

Type: Official workshop metadata identifying GridMind authors and forum

Full source URL: https://github.com/KDD26SciSocLLM/KDD26SciSocLLM.github.io/blob/master/accepted-papers.md

[PA04] Power-Agent/PowerMCP

Authors / issuer: PowerAgent community contributors

Publisher / venue: Author-maintained GitHub repository

Date: Live source repository reviewed 17 October 2026

Evidence type: Primary implementation and integration documentation

Full source URL: https://github.com/Power-Agent/PowerMCP

Visual credit: real_framework_nvidia_interaction_patterns.svg

Caption: Original NVIDIA interaction patterns. The same generative model can be part of a chat or an agentic system.

Credit: Source: NVIDIA Technical Blog, Figure 1.

Used on: 2. What is agentic AI?

Transformation: Only blank outer margins trimmed by display viewport; original image bytes are embedded unchanged and also retained as a separate original file.

Evidence limit: The three examples show interaction patterns, not a universal taxonomy or a claim that tool use alone proves autonomy.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://developer.nvidia.com/blog/building-for-the-rising-complexity-of-agentic-systems-with-extreme-co-design/

Original figure / image URL: https://developer-blogs.nvidia.com/wp-content/uploads/2026/05/image7.webp

Visual credit: real_concept_classical_agent.png

Caption: Russell and Norvig's original agent diagram, Figure 2.1. A software agent's sensors and actuators can be software interfaces.

Credit: Stuart J. Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, second edition, Chapter 2, Figure 2.1, printed p.33; author-hosted chapter draft.

Used on: 4. What is an AI agent?

Asset type: Original published textbook figure, native PDF crop

Transformation: Native PDF page clipped around Figure 2.1; no figure labels or connections redrawn.

Evidence limit: A general agent model, not a utility deployment or an LLM-specific architecture.

Original source / page: https://aima.cs.berkeley.edu/4th-ed/pdfs/newchap02.pdf

Original figure / image URL: https://aima.cs.berkeley.edu/4th-ed/pdfs/newchap02.pdf

Visual credit: real_framework_nvidia_agent_components.svg

Caption: Original NVIDIA agent design: the core connects a request with tools, planning and memory. Other designs can combine these roles.

Credit: Source: NVIDIA Technical Blog, Figure 1.

Used on: 5. An agent chooses a step. A tool runs it.

Transformation: Only blank outer margins trimmed by display viewport; original image bytes are embedded unchanged and also retained as a separate original file.

Evidence limit: This is a published conceptual design; separate memory or planning modules are not required in every AI agent.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://developer.nvidia.com/blog/introduction-to-llm-agents/

Original figure / image URL: https://developer-blogs.nvidia.com/wp-content/uploads/2023/11/agent-components.png

Visual credit: real_framework_langgraph_workflow_agent.svg

Caption: Original LangGraph documentation figure. The model chooses an action, receives feedback and can choose again.

Credit: Source: LangChain documentation maintainers, official LangGraph documentation. Accessed October 18, 2026.

Used on: 6. Workflow or agent: who chooses the next step?

Transformation: Selected original figure panel excerpt; no labels, arrows or pixels edited. Entire original image retained separately.

Evidence limit: This documentation convention concerns LLM applications, not all classical AI agents.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://docs.langchain.com/oss/python/langgraph/workflows-agents

Original figure / image URL: https://mintcdn.com/langchain-5e9cc07a/-_xGPoyjhyiDWTPJ/oss/images/agent_workflow.png

Visual credit: real_framework_langgraph_review_detail.svg

Caption: Original LangGraph Studio review-form excerpt. This is a generic demo, not a utility interface.

Credit: Source: LangChain documentation maintainers, official LangGraph documentation. Accessed October 18, 2026.

Used on: 7. An agent needs a clear goal and limits

Transformation: Viewport-only excerpt of the original approval form. Original image bytes embedded unchanged. Surrounding request and unused page whitespace excluded; no interface values edited.

Evidence limit: This is a generic software demonstration, not an operator dispatch interface or a utility deployment.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://docs.langchain.com/oss/python/langgraph/interrupts

Original figure / image URL: https://mintcdn.com/langchain-5e9cc07a/u62GP5wl5NeQqwjE/images/langgraph-interrupt-response-schema.png

Visual credit: source_anthropic_augmented_llm_wide.svg

Caption: Original Anthropic building-block figure. It is a general pattern, not a diagram of PowerAgent or PowerDAG.

Credit: Source: Anthropic, Building effective agents (2024), The augmented LLM. Original published figure, original diagram pixels unchanged; white margins trimmed for display.

Used on: 8. A model can use records, tools and memory

Transformation: White-margin-only display trim: SVG viewport clips an embedded byte-identical original PNG; PDF places original image samples on a cropped page. No diagram pixels or labels edited. The one-pixel outer frame is excluded.

Evidence limit: Published conceptual pattern, not an implementation diagram of PowerAgent, PowerDAG or a utility deployment.

License or permission status: No explicit image license verified; attribution is supplied, not a claim of an open license.

Original source / page: https://www.anthropic.com/engineering/building-effective-agents

Original figure / image URL: https://www-cdn.anthropic.com/images/4zrzovbb/website/d3083d3f40bb2b6f477901cc9a240738d3dd1371-2401x1000.png

Visual credit: real_news_pjm_hyperq_header.png

Caption: Original Tapestry article and published animation still. Supplier report, June 2026.

Credit: Tapestry, HyperQ Field Notes, June 11, 2026

Used on: 9. PJM uses HyperQ to review connection applications

Asset type: Genuine browser screenshot of original official source page

Transformation: Viewport crop of the live original webpage only. No text reconstruction, added labels, retouching, content changes or synthetic card.

Evidence limit: Supplier report of deployment. PJM separately confirmed earlier testing. The hero is a still captured from the publisher's animation, not an independently verified live PJM application record.

License or permission status: Original publisher content. No explicit open-reuse license independently verified. Attribution records the source and does not assert permission.

Original source / page: https://www.tapestryenergy.com/en/projects/field-notes-hyperq-deployment-in-pjm-s-cycle-1

Visual credit: power_news_hyperq_dashboard.png

Caption: Original Tapestry performance graphic, June 2026. The timing covers only computer processing.

Credit: Tapestry, HyperQ field notes, June 11, 2026

Used on: 10. HyperQ: what the reported numbers measure

Asset type: Original publisher site-control performance infographic; not an application-report screenshot

Transformation: None: original static PNG bytes, unmodified; no cropping, resizing, retouching, or GIF-frame conversion.

Evidence limit: Supplier-reported deployment metrics. Processing time excludes document transmission and human validation. 9,312 macro-assessments are not 9,312 applications.

Original source / page: https://www.tapestryenergy.com/en/projects/field-notes-hyperq-deployment-in-pjm-s-cycle-1

Original figure / image URL: https://cdn.sanity.io/images/bhywzdze/production/ee934ed00f306c6f3c2d0b5a92ab78dcb6278584-1257x651.png

Visual credit: caiso_control_room.jpg

Caption: Real California ISO control room. OATI supplies the batch result and later production-use claim.

Credit: California ISO control room, image published by OATI

Used on: 11. CAISO/OATI: agents help review planned outages

Asset type: Photograph

Transformation: None

Evidence limit: Control-room context photograph. The photo does not establish that any pictured display shows an AI agent or that an agent controls grid equipment.

License or permission status: Official OATI case-study page identifies this as the California ISO control room. No image-specific photographer or open reuse license stated. Permission not independently obtained.

Original source / page: https://www.oati.com/ai-genie/

Original figure / image URL: https://www.oati.com/wp-content/uploads/FolsomControlRoom5x7-1-1-2000x1435.jpg

Visual credit: duke_operator.jpg

Caption: Duke operator, 2022, shown for context. The reported agents support planning, rather than this pictured operating desk.

Credit: Duke Energy, illumination, August 2, 2022

Used on: 12. Duke and AWS: agents help prepare planning studies

Asset type: Photograph

Transformation: None

Evidence limit: Historical operator photograph used for operating context. The 2022 story does not document agentic AI.

License or permission status: Duke Energy Corporation, all rights reserved according to source-page footer. Permission not independently obtained.

Original source / page: https://illumination.duke-energy.com/articles/meet-the-team-that-monitors-the-grid-and-keeps-the-lights-on

Original figure / image URL: https://iprsoftwaremedia.com/264/files/20227/2022-0802-ECC-Greene-5.jpg

Visual credit: real__news__siemens__gridscale__header.png

Caption: Original Siemens press-release header, 2026. This is an announced product capability.

Credit: Siemens AG, press release, May 5, 2026

Used on: 13. Siemens announces agents for PSS E planning studies

Asset type: Faithful rendering of original official PDF page

Transformation: Original PDF page rendered by Poppler at aspect-preserving 1800px maximum dimension. No content edited. Source-preserving rectangular pixel crop [0, 0, 1273, 485]. Original text/image pixels unchanged; omitted surrounding page body, form or cookie-consent areas.

Evidence limit: Product announcement. The separately named Alliander platform migration does not prove production use of the new agentic PSS E capability.

License or permission status: Original publisher content. No explicit open-reuse license independently verified. Attribution records the source and does not assert permission.

Original source / page: https://press.siemens.com/global/en/pressrelease/siemens-gridscale-x-redefines-system-operations-and-agentic-transmission-planning

Original figure / image URL: https://assets.new.siemens.com/siemens/assets/api/uuid:fb613ac9-42b5-4e56-85be-0881ab9e95b8/HQSIPR202604177385EN.pdf

Visual credit: real__news__nypa__webinar__detail.svg

Caption: Original Ataccama webinar artwork. The NYPA speaker appears at lower right.

Credit: Ataccama, original webinar report with NYPA Data Quality Lead Saptaparna Gupta, October 14, 2026

Used on: 15. NYPA: better data for agent use

Asset type: Genuine browser screenshot of original official source page

Transformation: Viewport-only excerpt of the original webinar artwork. Page heading/date remain in the retained original screenshot; no artwork text or portraits changed.

Evidence limit: NYPA discusses meter-data quality checks. The separate product/reorder demo does not establish a production NYPA grid-control agent.

License or permission status: Original publisher content. No explicit open-reuse license independently verified. Attribution records the source and does not assert permission.

Original source / page: https://www.ataccama.com/webinar/on-demand-trust-driven-medallion-architecture

Visual credit: real__news__eversource__job__header.png

Caption: Original Eversource vacancy, 5 October 2026.

Credit: Eversource Energy, official engineering vacancy, October 5, 2026

Used on: 16. Eversource: adding agentic skills to grid studies

Asset type: Genuine browser screenshot of original official source page

Transformation: Viewport crop of the live original webpage only. No text reconstruction, added labels, retouching, content changes or synthetic card. Source-preserving rectangular pixel crop [0, 0, 1440, 875]. Original text/image pixels unchanged; omitted surrounding page body, form or cookie-consent areas.

Evidence limit: Engineering skill requirement and integration intent. The posting later explicitly mentions agentic MATLAB workflows, but a vacancy does not prove deployment.

License or permission status: Original publisher content. No explicit open-reuse license independently verified. Attribution records the source and does not assert permission.

Original source / page: https://jobs.eversource.com/job/east-berlin/distribution-system-modeling-and-automation-engineer-hybrid-schedule/7774/101576695504

Visual credit: real__news__eversource__agentic__line.svg

Caption: Original requirement excerpt. Hiring and integration intent; deployment is not established.

Credit: Eversource Energy, official engineering vacancy, October 5, 2026

Used on: 16. Eversource: adding agentic skills to grid studies

Asset type: Genuine browser element screenshot of original published job requirements

Transformation: Viewport selects the single complete agentic-workflow requirement from the original source-element screenshot. Original image bytes retained unchanged.

Evidence limit: Advertised engineering skills. This is hiring/integration intent, not proof of a production agent platform.

License or permission status: Original publisher content. No explicit open-reuse license independently verified. Attribution records the source and does not assert permission.

Original source / page: https://jobs.eversource.com/job/east-berlin/distribution-system-modeling-and-automation-engineer-hybrid-schedule/7774/101576695504

Visual credit: ge_gridos_control_room.jpg

Caption: Official GridOS promotional composite. The platform example does not establish a measured agent-specific result.

Credit: GE Vernova

Used on: 17. GE Vernova: a platform that can connect grid tools

Asset type: Official product promotional composite

Transformation: Format conversion from WebP to JPEG only

Evidence limit: Promotional product imagery, not evidence that an autonomous agent operates a named U.S. utility control room.

License or permission status: Official GE Vernova product-page image. No explicit reuse license found. Permission not independently obtained.

Original source / page: https://www.gevernova.com/software/products/gridos

Original figure / image URL: https://www.gevernova.com/software/sites/default/files/2026-06/2024-grid-gridos-hero-image-1600x536.webp

Visual credit: duke_control_room.jpg

Caption: Duke Energy control room, 2022. The PNNL survey is a separate study.

Credit: Duke Energy, illumination, August 2, 2022

Used on: 18. Operators often work against a deadline

Asset type: Photograph

Transformation: None

Evidence limit: Historical operator photograph used for operating context. The 2022 story does not document agentic AI.

License or permission status: Duke Energy Corporation, all rights reserved according to source-page footer. Permission not independently obtained.

Original source / page: https://illumination.duke-energy.com/articles/meet-the-team-that-monitors-the-grid-and-keeps-the-lights-on

Original figure / image URL: https://iprsoftwaremedia.com/264/files/20227/62e806b2b3aed342060cf206_2022-0802-ECC-Greene-hero/2022-0802-ECC-Greene-hero_hero.jpg?v=b4580dcf-c141-4a6a-b29f-a5038a74cb1a

Visual credit: caiso_watch_duty.png

Caption: Real CAISO Watch Duty display. It combines fire and transmission-line information for operator awareness.

Credit: California ISO. Application: Watch Duty.

Used on: 19. Three checks for a useful recommendation

Asset type: Actual operational situational-awareness tool screenshot

Transformation: Format conversion from WebP to PNG only

Evidence limit: The source describes operators overlaying transmission lines on fire maps and reviewing webcam images to assess fire proximity. This screenshot is not agentic AI and does not establish autonomous control. Native source resolution is modest, so avoid large enlargement.

License or permission status: Official California ISO control-center article screenshot. Source footer states all rights reserved. No image-specific reuse license found. Permission not independently obtained.

Original source / page: https://www.caiso.com/about/news/energy-matters-blog/control-center-modernization-is-enhancing-reliability

Original figure / image URL: https://www.caiso.com/content/images/blog/blog-graphics/controlcentermodernization-watchduty.webp

Visual credit: ge_gridos_visual_intelligence.png

Caption: GE Vernova's published GridOS Visual Intelligence screen. This product image supplies task context, not an agent result.

Credit: GE Vernova

Used on: 20. A clear task makes success easier to test

Asset type: Official product screenshot and video thumbnail

Transformation: None

Evidence limit: Illustrates vendor software, not a named U.S. agentic AI deployment.

License or permission status: Official GE Vernova product-page video thumbnail. No explicit reuse license found. Permission not independently obtained.

Original source / page: https://www.gevernova.com/software/products/gridos/visual-intelligence

Original figure / image URL: https://www.gevernova.com/software/sites/default/files/2026-10/frame-1171277779.png

Visual credit: source_anthropic_autonomous_agent_wide.svg

Caption: Original Anthropic agent-loop figure. In a grid study, feedback can be solver results or a failed check.

Credit: Source: Anthropic, Building effective agents (2024), Autonomous agent. Original published figure, original diagram pixels unchanged; white margins trimmed for display.

Used on: 21. An LLM agent acts, reads the result and decides again

Transformation: White-margin-only display trim: SVG viewport clips an embedded byte-identical original PNG; PDF places original image samples on a cropped page. No diagram pixels or labels edited. The one-pixel outer frame is excluded.

Evidence limit: Published conceptual pattern, not an implementation diagram of PowerAgent, PowerDAG or a utility deployment.

License or permission status: No explicit image license verified; attribution is supplied, not a claim of an open license.

Original source / page: https://www.anthropic.com/engineering/building-effective-agents

Original figure / image URL: https://www-cdn.anthropic.com/images/4zrzovbb/website/58d9f10c985c4eb5d53798dea315f7bb5ab6249e-2401x1000.png

Visual credit: real_framework_anthropic_prompt_chain.svg

Caption: Original Anthropic gated workflow. In our study example, a stale input should fail the check and stop the sequence.

Credit: Source: Anthropic, Building effective agents (December 19, 2024).

Used on: 22. When data are stale, the next step is to stop

Transformation: Only blank outer margins trimmed by display viewport; original image bytes are embedded unchanged and also retained as a separate original file.

Evidence limit: A generic content-workflow pattern, not a physical-feasibility or utility-safety guarantee.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://www.anthropic.com/engineering/building-effective-agents

Original figure / image URL: https://www-cdn.anthropic.com/images/4zrzovbb/website/7418719e3dab222dccb379b8879e1dc08ad34c78-2401x1000.png

Visual credit: real_framework_anthropic_routing.svg

Caption: Original Anthropic routing pattern. Choosing an engineering study also requires selecting the right problem and tool.

Credit: Source: Anthropic, Building effective agents (December 19, 2024).

Used on: 25. A solver result depends on the problem it solves

Transformation: Only blank outer margins trimmed by display viewport; original image bytes are embedded unchanged and also retained as a separate original file.

Evidence limit: The picture illustrates routing; it does not establish an implemented power-system workflow.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://www.anthropic.com/engineering/building-effective-agents

Original figure / image URL: https://www-cdn.anthropic.com/images/4zrzovbb/website/5c0c0e9fe4def0b584c04d37849941da55e5e71c-2401x1000.png

Visual credit: epri__open__power__ai__grid.png

Caption: Image accompanying the EPRI Open Power AI Consortium announcement, published by NVIDIA, 20 March 2025.

Credit: Image published by NVIDIA with its EPRI Open Power AI Consortium announcement, March 20, 2025

Used on: 28. Power-system papers test different tasks

Asset type: Photograph

Transformation: None

Evidence limit: Contextual grid photograph. The pictured location and utility are not identified.

License or permission status: Official NVIDIA blog image. No photographer or explicit reuse license stated on the source page. Permission not independently obtained.

Original source / page: https://blogs.nvidia.com/blog/open-power-ai-consortium/

Original figure / image URL: https://blogs.nvidia.com/wp-content/uploads/2025/03/cudss-featured-1280x680-blog.png

Visual credit: real__paper__poweragent__three__layers.png

Caption: Original Harvard author roadmap: workflows, tool connections and foundation models. Three-layer excerpt.

Credit: Le Xie; slides credit Le Xie and Qian Zhang; A Roadmap Towards Agentic Intelligence in Power Systems; Author presentation slide 11

Used on: 29. Harvard PowerAgent: models, tools and study workflows

Asset type: Original published figure or result panel, native PDF crop

Transformation: Native PDF crop selects the three layer panels from the author roadmap. No text or connectors redrawn; the full source figure is retained.

Evidence limit: Roadmap from the December2025 author presentation; not a measured deployment or a frozen specification of every current PowerAgent repository.

Original source / page: https://www.ercot.com/files/docs/2025/12/01/4-Guest-Presentation-A-Roadmap-toward-Agentic-Intelligence-in-Power-Systems.pdf

Original figure / image URL: https://www.ercot.com/files/docs/2025/12/01/4-Guest-Presentation-A-Roadmap-toward-Agentic-Intelligence-in-Power-Systems.pdf

Visual credit: real_paper_poweragent_impact_workflow.png

Caption: Original Harvard concept for grid-impact studies. The separate PowerWF example uses LangGraph and PowerWorld; this is not its exact graph.

Credit: Le Xie; slides credit Le Xie and Qian Zhang; A Roadmap Towards Agentic Intelligence in Power Systems; Author presentation slide 14, right-hand grid-impact process

Used on: 30. PowerAgent example: what changes when load is added?

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: A conceptual author demonstration. It is not an exact diagram of the later PowerWF graph.py or a utility performance trial.

Original source / page: https://www.ercot.com/files/docs/2025/12/01/4-Guest-Presentation-A-Roadmap-toward-Agentic-Intelligence-in-Power-Systems.pdf

Original figure / image URL: https://www.ercot.com/files/docs/2025/12/01/4-Guest-Presentation-A-Roadmap-toward-Agentic-Intelligence-in-Power-Systems.pdf

Visual credit: power_dragent_waveform.png

Caption: Original DrAgent Figure 2, PDF p.4. The third current channel shows a fault in a sample disturbance record.

Credit: Saha, V and Naidu / Hitachi Energy, DrAgent, ICAIIC 2025, Figure 2

Used on: 31. DrAgent: reading a recorded grid fault

Asset type: Original scientific figure, native PDF crop

Transformation: Clipped to the original figure boundary; caption excluded and credited separately; no plotted data changed.

Evidence limit: A sample disturbance record from a research paper; not a live utility operating display.

Original source / page: https://www.manuscriptlink.com/society/kics/media?key=kics%2Fconference%2Ficaiic2025%2Fpresentation%2F1571098105.pdf

Original figure / image URL: https://www.manuscriptlink.com/society/kics/media?key=kics%2Fconference%2Ficaiic2025%2Fpresentation%2F1571098105.pdf

Visual credit: real_paper_dragent_architecture.png

Caption: Original DrAgent architecture: planning, tool execution and answer generation.

Credit: Barun Kumar Saha, Aarthi V, O. D. Naidu; DrAgent: An Agentic Approach to Fault Analysis in Power Grids Using Large Language Models; Figure 1, PDF page 2

Used on: 32. DrAgent selects tools to answer a fault question

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: Paper architecture based on ReWOO; the picture does not establish utility deployment or autonomous command authority.

Original source / page: https://www.manuscriptlink.com/society/kics/media?key=kics%2Fconference%2Ficaiic2025%2Fpresentation%2F1571098105.pdf

Original figure / image URL: https://www.manuscriptlink.com/society/kics/media?key=kics%2Fconference%2Ficaiic2025%2Fpresentation%2F1571098105.pdf

Visual credit: power_dragent_counts.png

Caption: Counts reconstructed from Table IV and the Appendix. They are neither new experiments nor the paper's mean-score measure.

Credit: Original educational plot derived from published DrAgent score cells; no experiments rerun.

Used on: 33. DrAgent: more correct answers, with errors still present

Asset type: Original quantitative teaching figure

Provenance note: Correct answers (score >=2), reconstructed from Table IV and Appendix; n=18 DR cases per question. Q6 includes correctly labeled duration under footnote 8. Counts are author-data-derived, not an independent rerun.

Evidence limit: 18 disturbance-record cases, not 18 repeated stochastic trials per case. Scores were manually assessed by a domain expert. Baseline and agent use different prompt/context workflows. The comparison does not isolate one component causally. Do not describe the Q6 result as 13/18 correct clearance timestamps. Q6 printed mean is inconsistent; audited cell counts are used instead.

Original source / page: https://www.manuscriptlink.com/society/kics/media?key=kics%2Fconference%2Ficaiic2025%2Fpresentation%2F1571098105.pdf

Original figure / image URL: images/power_dragent_counts.png

Visual credit: real_framework_langgraph_static_interrupt.svg

Caption: Real LangGraph debugger screenshot. A paused run can be inspected before the next software step.

Credit: Source: LangChain documentation maintainers, official LangGraph documentation. Accessed October 18, 2026.

Used on: 36. LangGraph can pause to inspect a run

Transformation: Only blank outer margins trimmed by display viewport; original image bytes are embedded unchanged and also retained as a separate original file.

Evidence limit: A static debug breakpoint differs from a dynamic interrupt requesting human input.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://docs.langchain.com/oss/python/langgraph/interrupts

Original figure / image URL: https://mintcdn.com/langchain-5e9cc07a/dL5Sn6Cmy9pwtY0V/oss/images/static-interrupt.png

Visual credit: real_framework_langgraph_state.svg

Caption: Original LangGraph state diagram. Software checkpoints still need grid-case, time and evidence identifiers.

Credit: Source: LangChain documentation maintainers, official LangGraph documentation. Accessed October 18, 2026.

Used on: 37. Saved state must identify the study and its evidence

Transformation: Selected original figure panel excerpt; no labels, arrows or pixels edited. Entire original image retained separately.

Evidence limit: The extra grid identifiers are tutorial requirements; they are not fields proven by this generic illustration.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://docs.langchain.com/oss/python/langgraph/checkpointers

Original figure / image URL: https://mintcdn.com/langchain-5e9cc07a/-_xGPoyjhyiDWTPJ/oss/images/get_state.jpg

Visual credit: real_framework_langgraph_checkpoint.svg

Caption: Original LangGraph checkpoint diagram. Saving progress does not prove that the grid data still match.

Credit: Source: LangChain documentation maintainers, official LangGraph documentation. Accessed October 18, 2026.

Used on: 38. A checkpoint lets a workflow pause and resume

Transformation: Selected original figure panel excerpt; no labels, arrows or pixels edited. Entire original image retained separately.

Evidence limit: Saved software state does not prove that an external grid snapshot remains current.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://docs.langchain.com/oss/python/langgraph/checkpointers

Original figure / image URL: https://mintcdn.com/langchain-5e9cc07a/-_xGPoyjhyiDWTPJ/oss/images/checkpoints.jpg

Visual credit: real__framework__anthropic__parallel.svg

Caption: Original Anthropic parallel workflow. A grid study must also account for missing or failed outage cases.

Credit: Source: Anthropic, Building effective agents (December 19, 2024).

Used on: 39. Parallel outage studies must account for every case

Transformation: Only blank outer margins trimmed by display viewport; original image bytes are embedded unchanged and also retained as a separate original file.

Evidence limit: The aggregator picture does not imply that missing results are automatically detected.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://www.anthropic.com/engineering/building-effective-agents

Original figure / image URL: https://www-cdn.anthropic.com/images/4zrzovbb/website/406bb032ca007fd1624f261af717d70e6ca86286-2401x1000.png

Visual credit: real__paper__powerdag__architecture.png

Caption: Original PowerDAG v4 architecture, including the supervisor before environment execution.

Credit: Emmanuel O. Badmus, Amritanshu Pandey; PowerDAG: Supervisory Agentic AI System for Automating Distribution Grid Analysis; Figure 2, v4 PDF page 3

Used on: 40. PowerDAG checks the steps needed before a tool runs

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: This is the v4 research architecture. Formal supervision checks encoded prior Write-tool names; it does not establish all freshness, parameter or physical-safety checks.

Original source / page: https://arxiv.org/pdf/2603.17418v4

Original figure / image URL: https://arxiv.org/pdf/2603.17418v4

Visual credit: powerdag__revision__result.png

Caption: PowerDAG v4, Table II. ReAct: a reason-and-act loop. JIT: prerequisite checks just before a call.

Credit: Original vector plot redrawn from author-reported PowerDAG v4 numerical results

Used on: 42. PowerDAG results: the same queries, four setups

Asset type: Original quantitative teaching figure

Provenance note: PowerDAG v4 same-model comparison on150 held-out distribution-grid queries: GPT-5.5 ReAct77.33%, adaptive-retrieval-only92.67%, JIT-supervision-only96.00%, fullPowerDAG98.00%. Each query is run once per model/configuration; no experiments rerun.

Evidence limit: Pass@1 checks expert tool coverage, compatible arguments and dependency ordering. Four real-world feeder models and ten task families; within-family holdout, not field reliability. No run-to-run interval is inferred. Public network-dependent reruns require proprietary data/tools; only3of10families are fully runnable publicly.

Original source / page: https://arxiv.org/pdf/2603.17418v4

Original figure / image URL: images/powerdag_revision_result.png

Visual credit: real_paper_daline_feedback.png

Caption: Original environmental acting module: simulation errors feed retrieval and code repair.

Credit: Mengshuo Jia, Zeyu Cui, Gabriela Hug; Enhancing LLMs for Power System Simulations: A Feedback-Driven Multi-Agent Framework; Figure 5, author-preprint PDF page 7

Used on: 44. Simulation agents find functions, run code and repair errors

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: Published workflow module, not evidence that error-free execution satisfies the requested engineering settings.

Original source / page: https://doi.org/10.1109/TSG.2025.3589114

Original figure / image URL: https://arxiv.org/pdf/2411.16707v3

Visual credit: power_study_feedback.png

Caption: DALINE comparison reported in the author preprint, Figure 8 / section V. Each task allows up to three attempts.

Credit: Original plot redrawn from cited primary-paper numerical results; no experiments rerun

Used on: 45. Simulation-agent results: points, not a pass rate

Asset type: Original quantitative teaching figure

Provenance note: DALINE graded attempt-averaged scores: GPT4o-Sole 0.00, GPT4o-SR 31.37, CGPT4o-R 33.82 and GPT4o-Full 93.13. The evaluation uses 34 tasks with a maximum of three attempts each, awarding 100/50/0 points and carrying the last executed score into unused slots. This is not a final-completion fraction.

Evidence limit: Reported values preserved. API schemes use gpt-4o-2024-05-13; CGPT4o-R is the ChatGPT4o built-in-RAG product scheme. Complete-system comparison, not isolated causal effects. No binomial interval or implied completed-task count.

Original source / page: https://arxiv.org/pdf/2411.16707v3

Original figure / image URL: images/power_study_feedback.png

Visual credit: real_framework_anthropic_evaluator.svg

Caption: Original Anthropic feedback pattern. A retry must actually use the repaired code and changed inputs.

Credit: Source: Anthropic, Building effective agents (December 19, 2024).

Used on: 46. Trying again can repeat the same software error

Transformation: Only blank outer margins trimmed by display viewport; original image bytes are embedded unchanged and also retained as a separate original file.

Evidence limit: An LLM evaluator does not prove grid feasibility, permission or fresh input data.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://www.anthropic.com/engineering/building-effective-agents

Original figure / image URL: https://www-cdn.anthropic.com/images/4zrzovbb/website/14f51e6406ccb29e695da48b17017e899a6119c7-2401x1000.png

Visual credit: platform__pLPJoFvq4__M.jpg

Caption: LangChain / Launch walkthrough linked 1 August 2024

Credit: LangChain. LangSmith Studio: The first agent IDE. Launch walkthrough linked 1 August 2024. Official video thumbnail.

Used on: 47. LangGraph Studio: watch an agent use tools

Asset type: Official video thumbnail

Transformation: None

Evidence limit: General software-platform demonstration; not power-system validation or utility deployment evidence.

License or permission status: Publisher thumbnail linked to the official video; copyright remains with the publisher.

Original source / page: https://www.youtube.com/watch?v=pLPJoFvq4_M

Original figure / image URL: https://i.ytimg.com/vi/pLPJoFvq4_M/hqdefault.jpg

Visual credit: platform__682ca322-5d0b-47a0-8d18-77bf9d0fb016.png

Caption: Siemens / Undated video; accessed 26 October 2026

Credit: Siemens. Gridscale X PSS E: agentic transmission planning — product video. Undated video; accessed 26 October 2026. Official video thumbnail.

Used on: 48. Siemens PSS E: agentic transmission planning

Asset type: Official video thumbnail

Transformation: None

Evidence limit: General software-platform demonstration; not power-system validation or utility deployment evidence.

License or permission status: Publisher thumbnail linked to the official video; copyright remains with the publisher.

Original source / page: https://video.sw.siemens.com/embed/682ca322-5d0b-47a0-8d18-77bf9d0fb016

Original figure / image URL: https://video.sw.cdn.siemens.com/mgc/thumbnails/682ca322-5d0b-47a0-8d18-77bf9d0fb016-en-US-thumbnail.png

Visual credit: real__paper__gridmind2025__architecture.png

Caption: Original GridMind2025 diagram of specialized agents and numerical analysis tools.

Credit: Hongwei Jin, Kibaek Kim, Jonghwan Kwon; GridMind: LLMs-Powered Agents for Power System Analysis and Operations; Figure 2, PDF page 2

Used on: 49. GridMind 2025 connects typed agents to solvers

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: This is the2025 PydanticAI/pandapower study, not the separate2026 LangGraph weather-risk system.

Original source / page: https://arxiv.org/abs/2509.02494v1

Original figure / image URL: https://arxiv.org/pdf/2509.02494v1

Visual credit: real_paper_gridmind2025_response.png

Caption: A separate published response claims no thermal overloads and flags a balance error. It is not independent ground truth.

Credit: Hongwei Jin, Kibaek Kim, Jonghwan Kwon; GridMind: LLMs-Powered Agents for Power System Analysis and Operations; Figure 8, PDF page 10, baseline and coverage portion of agent response

Used on: 50. GridMind: reported answers need independent checks

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: This is an example response, not independent ground truth or the Table1 comparison. Its own output reports a power-balance issue; an answer still needs engineering review.

Original source / page: https://arxiv.org/abs/2509.02494v1

Original figure / image URL: https://arxiv.org/pdf/2509.02494v1

Visual credit: expanded_power_gridmind.png

Caption: GridMind 2026 Table 1: full reruns in fresh workspaces versus saved-result reuse, on a synthetic 2,000-bus hurricane-risk study.

Credit: Original chart from cited author-reported session/ablation measurements

Used on: 53. GridMind results: fewer stages and less time are different

Asset type: Original quantitative teaching figure

Provenance note: GridMind weather-risk workflow, same seven-query session on a synthetic Texas 2000-bus case. Full-rerun ablation executes 35 stages and takes 1432.6 s; content-addressed reuse executes 14 stages and takes 677.1 s. The stage-count reduction is 60%; the calculated wall-time speedup is 2.1158 times (2.12 times rounded), corresponding to 52.7363% less wall time. These are different metrics, not two descriptions of the same percentage. No repeated-session uncertainty estimate is reported.

Evidence limit: Same reported query sequence; full-rerun arm uses fresh namespaces. Single simulation session, surrogate screening and reduced-order cascade models. No live utility operation or independently repeated performance claim.

Original source / page: https://openreview.net/pdf/5ad1c2afd7298f9e6ce3bc25f18e7165555e1ffd.pdf

Original figure / image URL: images/expanded_power_gridmind.png

Visual credit: real__paper__dynamicbench__overview.png

Caption: Original benchmark structure: input package, agentic system, tool calls and evaluation metrics.

Credit: Qian Zhang, Andrea Pomarico, Costas Mylonas, Magda Foti, Alberto Berizzi, Le Xie; PowerAgentBench-Dyn: A Benchmark for Agentic AI in Power System Dynamic Studies; Figure 1, PDF page 2

Used on: 56. PowerAgentBench-Dyn allows only specific model changes

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: Benchmark architecture, not the full PowerAgent roadmap or an operating utility deployment.

Original source / page: https://arxiv.org/abs/2606.20401v1

Original figure / image URL: https://arxiv.org/pdf/2606.20401v1

Visual credit: expanded__power__dynamic.png

Caption: PowerAgentBench-Dyn Table II: model-quality task. Other tasks in this benchmark use different models and checks.

Credit: Original chart from cited author-reported research results

Used on: 57. PowerAgentBench-Dyn results: passed sessions and updates

Asset type: Original quantitative teaching figure

Provenance note: PowerAgentBench-Dyn DMQ case: ten independent sessions per model on one modified WECC solar PV dynamic model. Full-suite compliant success is 10/10 for Opus 4.8, 10/10 for Sonnet 4.6, and 9/10 for Haiku 4.5. Median updates are 1, 1 and 2. Eight quality tests and a maximum of five parameter updates define this case. No confidence interval or fleet-level performance is claimed.

Evidence limit: Author-reported counts and medians reproduced, not independently rerun. One fixed corrupted model; no human or deterministic search comparator.

Original source / page: https://arxiv.org/abs/2606.20401v1

Original figure / image URL: images/expanded_power_dynamic.png

Visual credit: real__paper__veragrid__agent__workflow.png

Caption: Original VeraGrid-Agent workflow: question, LLM controller, input file, AC-OPF workspace and selected result reads.

Credit: Shivanshu Tripathi, Hamed Mohsenian-Rad, Maziar Raissi; VeraGrid-Agent: Tool-Augmented LLMs for Distribution Optimal Power Flow at the Grid Edge; Figure 2, PDF page 2

Used on: 59. VeraGrid-Agent uses solver results to answer a question

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: A research question-answering workflow on one feeder. It is not a general feeder-control policy.

Original source / page: https://arxiv.org/pdf/2607.25155v1

Original figure / image URL: https://arxiv.org/pdf/2607.25155v1

Visual credit: power_study_veragrid.png

Caption: VeraGrid-Agent Table I: selected comparisons using the same model. Missing or badly formatted answers count as incorrect.

Credit: Original plot redrawn from cited primary-paper numerical results; no experiments rerun

Used on: 60. VeraGrid-Agent results: the same models with and without tools

Asset type: Original quantitative teaching figure

Provenance note: Three explicitly selected same-model comparisons from VeraGrid-MCQ-150: GPT-5.2 64/150 without tools versus 149/150 with the solver-agent; Claude Sonnet 4.5 71/150 versus 149/150; Gemini 3 Flash 74/150 versus 150/150. Questions use one balanced 8-bus feeder and a VeraGrid reference solve.

Evidence limit: First three Table I rows selected for legibility, not ranked. Template-derived MCQs share one solved feeder. Repeated-run uncertainty is unspecified; results do not estimate field-control reliability or arbitrary-feeder generalization.

Original source / page: https://arxiv.org/pdf/2607.25155v1

Original figure / image URL: images/power_study_veragrid.png

Visual credit: real_paper_agentigrid_architecture.png

Caption: Original AgentiGrid data flow: loop controller, analysis-controller interface and ExaGO application.

Credit: Samim Konjicija, Slaven Peles; Integrating Agentic Artificial Intelligence with High-Performance Computing for Grid Planning; Figure 1, PDF page 2

Used on: 61. AgentiGrid: what does one reported run mean?

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: Prototype architecture for modeled studies. Twelve reported searches are different backend/application settings, not twelve repeats of one task.

Original source / page: https://arxiv.org/abs/2609.04544v1

Original figure / image URL: https://arxiv.org/pdf/2609.04544v1

Visual credit: rigor_r01_coverage.png

Caption: Counts read from R01 Table III and section IV. Each outage label is a pair of bus numbers in this test grid.

Credit: Original plot derived from cited paper data; see caption for interpretation

Used on: 63. Successful tool calls can still miss a critical outage

Asset type: Original quantitative teaching figure

Provenance note: Derived from R01 Table III: the four reported backend searches on one synthetic ACTIVSg200 SCOPFLOW case all found economic-critical contingency 187–189 (+21.6% cost), three identified infeasible outage 29–30, and none identified infeasible outage 15–16. These are fixed backend-run counts, not repeated IID trials, estimated probabilities, or comprehensive field reliability. No confidence interval is justified by these four heterogeneous searches.

Evidence limit: One synthetic case, four different backend searches, reported search behavior only.

Original source / page: https://arxiv.org/pdf/2609.04544v1

Original figure / image URL: images/rigor_r01_coverage.png

Visual credit: rigor_r02_capability.png

Caption: R02 Table II. L1: place a 100 MW + 10 Mvar load. L2: check reactive-power needs for 100 MW + 50 Mvar generation.

Credit: Original plot derived from cited paper data; see caption for interpretation

Used on: 64. Show both trial counts and uncertainty

Asset type: Original quantitative teaching figure

Provenance note: Derived from R02 Table II, PDF page 4, tested development snapshot. Bars reproduce the eight reported 100/1000-bus count cells. Whiskers are calculated two-sided Wilson 95% intervals, not author-reported uncertainty. Their probabilistic interpretation requires comparable independent Bernoulli attempts, which the paper does not establish. The OpenAI table label is gpt 5.4 B. Larger grids and levels L3–L6 have no quantified result cells and are omitted, not encoded as zero. This is not field-deployment reliability or a current vendor ranking.

Evidence limit: Reported counts preserved. Wilson intervals conditional on independent comparable trials. Model/backend and task snapshot specific.

Original source / page: https://arxiv.org/pdf/2607.28822v1

Original figure / image URL: images/rigor_r02_capability.png

Visual credit: real__paper__powerbench__dataset.png

Caption: Original PowerBench overview linking structured asset state, telemetry, rules, records and queries.

Credit: Xijing Wang, Yinsheng Yao, Jinru Ding, Yidong Jiang, Ziwen Xu, Yiwen Jiang, Jie Xu, Dawei Cheng; PowerBench: A Benchmark for Agentic Retrieval and Reasoning in Power Systems; Figure 1, PDF page 4

Used on: 65. PowerBench finds answers across linked records

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: Synthetic benchmark data, not utility operational records, simulation results or a live grid database.

Original source / page: https://arxiv.org/abs/2609.34492v1

Original figure / image URL: https://arxiv.org/pdf/2609.34492v1

Visual credit: expanded__power__benchmark.png

Caption: PowerBench Table 4: selected models and the authors' 95% bootstrap intervals. The two metrics average results differently.

Credit: Original chart from cited author-reported scores and bootstrap intervals

Used on: 66. PowerBench results: complete answers and partial credit

Asset type: Original quantitative teaching figure

Provenance note: Selected PowerBench Table 4 results. Macro Joint Accuracy equally weights the three task families, with 60 temporal, 90 contextual and 150 comparative questions; it is not a correct-count divided by 300. Overall Field Accuracy is the mean per-question fraction of correct fields. Whiskers are author-reported 95% bootstrap intervals, not newly calculated intervals. All three configurations used the same 300-question benchmark and 50-call/300-second retrieval-and-reasoning budgets.

Evidence limit: Different aggregation rules are retained. Partial field credit is not full-task success. Do not infer pairwise significance from marginal interval overlap. Benchmark results are synthetic and author-reported; public runnable release not verified.

Original source / page: https://arxiv.org/abs/2609.34492v1

Original figure / image URL: images/expanded_power_benchmark.png

Visual credit: real__paper__powerbench__tasks.png

Caption: Original PowerBench task examples. Each task group contributes equally to the reported average.

Credit: Xijing Wang, Yinsheng Yao, Jinru Ding, Yidong Jiang, Ziwen Xu, Yiwen Jiang, Jie Xu, Dawei Cheng; PowerBench: A Benchmark for Agentic Retrieval and Reasoning in Power Systems; Figure 2, PDF page 6

Used on: 67. A strong average can hide a weak task group

Asset type: Original published figure or result panel, native PDF crop

Transformation: Only the stated page rectangle is retained; original labels, plotted content and colors are unchanged. Non-figure page-link annotations are removed. Native PDF content is preserved, with a 240-dpi PNG for web display.

Evidence limit: Illustrates three task families; the metric gives them equal weight even though their sizes are60,90 and150 questions.

Original source / page: https://arxiv.org/abs/2609.34492v1

Original figure / image URL: https://arxiv.org/pdf/2609.34492v1

Visual credit: rigor__reliability__chain.png

Caption: Illustrative calculation, not measured agent performance: P(all L steps succeed) = q^L.

Credit: Original illustrative calculation for this tutorial; not measured agent performance

Used on: 69. More required steps can lower overall reliability

Asset type: Original quantitative teaching figure

Provenance note: Illustrative calculation: if every step has identical success probability q and steps are independent, all L required steps succeed with probability q^L. These are mathematical scenarios, not measured agent reliability. Correlated errors or unequal steps require a different model.

Evidence limit: Identical independent Bernoulli step outcomes. All required steps must succeed.

Original figure / image URL: images/rigor_reliability_chain.png

Visual credit: rigor__eval__outcomes.png

Caption: Invented example: 80 feasible and 20 infeasible cases. These counts explain metrics; they are not deployment results.

Credit: Original illustrative calculation for this tutorial; not measured agent performance

Used on: 70. The answer changes when we change what we count

Asset type: Original quantitative teaching figure

Provenance note: Hypothetical evaluation of 100 cases, not measured system performance. Of 80 truly feasible cases, 70 are accepted and 10 rejected. Of 20 truly infeasible cases, 2 are accepted and 18 rejected. False approval among infeasible cases is 2/20=10%, while the pooled fraction is 2/100=2%. Accepted-case precision is 70/72=97.22%. The denominator answers a different question in each metric.

Evidence limit: Illustrative labels and outcomes, fixed failure definition. No measured agent data.

Original figure / image URL: images/rigor_eval_outcomes.png

Visual credit: rigor_zero_failure_bound.png

Caption: Calculated binomial bound. It needs a fixed failure definition, independent trials and stable test conditions.

Credit: Original illustrative calculation for this tutorial; not measured agent performance

Used on: 71. Zero test failures do not prove perfect performance

Asset type: Original quantitative teaching figure

Provenance note: Exact one-sided 95% binomial upper confidence bound after zero failures: p_upper = 1 - 0.05^(1/n). At n=30 the bound is 9.503%, at n=60 it is 4.8703%, at n=300 it is 0.9936%, and at n=3000 it is 0.09981%. It describes uncertainty under independent, identically distributed trials with a fixed failure definition and stable operating conditions. It does not establish reliability under a new topology or a new task distribution.

Evidence limit: IID binomial trials, zero observed failures, prespecified one-sided 95% confidence procedure. This bound is calculated, not author-reported.

Original figure / image URL: images/rigor_zero_failure_bound.png

Visual credit: real_framework_mcp_architecture.png

Caption: Original MCP architecture. Clients connect to servers; the engineering tool performs the calculation.

Credit: Source: Model Context Protocol, official documentation revision July 28, 2026. Screenshot captured October 18, 2026.

Used on: 73. MCP connects a workflow to its tools

Transformation: Native browser screenshot of the original rendered source diagram only. Page navigation and surrounding prose excluded; diagram content unchanged.

Evidence limit: A protocol architecture does not prove numerical correctness or that a utility uses this implementation.

License or permission status: Publisher attribution supplied. No separate permission or unrestricted reuse license claimed.

Original source / page: https://modelcontextprotocol.io/docs/2026-07-28/learn/architecture

Original figure / image URL: https://modelcontextprotocol.io/docs/2026-07-28/learn/architecture

Visual credit: real_ops_frequency.png

Caption: Real CAISO oscillation-monitoring screens. The operator must still interpret the result and decide what to do.

Credit: California ISO, John Phipps, Control center modernization is enhancing reliability, 10 September 2025.

Used on: 75. Measure the whole job, including human review

Asset type: Original operational software screenshot

Transformation: Original image; format conversion only where required.

Evidence limit: Context photograph or conventional operator tool; no agent performance or control authority is implied.

Original source / page: https://www.caiso.com/about/news/energy-matters-blog/control-center-modernization-is-enhancing-reliability

Original figure / image URL: https://www.caiso.com/content/images/blog/blog-graphics/controlcentermodernization-rtdms.webp

Visual credit: real_ops_training.png

Caption: California ISO's operator-training facility. Photo: California ISO, 2025.

Credit: California ISO, Tricia Johnstone, ISO now has expanded training and emergency operations facilities, 20 February 2025.

Used on: 76. Test whether the assistant helps on comparable tasks

Asset type: Original utility training-facility photograph

Transformation: Original image; format conversion only where required.

Evidence limit: Context photograph or conventional operator tool; no agent performance or control authority is implied.

Original source / page: https://www.caiso.com/about/news/energy-matters-blog/iso-now-has-expanded-training-and-emergency-operations-facilities

Original figure / image URL: https://www.caiso.com/content/images/blog/blog-graphics/training2.webp

Visual credit: ornl_goal.jpg

Caption: ORNL GOAL research laboratory, shown for context. This is not a photograph of LLNL's testbed.

Credit: Oak Ridge National Laboratory, U.S. Department of Energy

Used on: 77. Power-sector testbeds provide places to evaluate agents

Asset type: Photograph

Transformation: None

Evidence limit: The source describes a research test bed and replicated utility command center. Do not label this as a production utility deployment.

License or permission status: Official ORNL laboratory photograph. No explicit image-specific reuse license found. ORNL is managed by UT-Battelle for DOE, so public-domain status is not assumed. Permission not independently obtained.

Original source / page: https://www.ornl.gov/grid-operations-and-analytics-laboratory-goal/

Original figure / image URL: https://www.ornl.gov/wp-content/uploads/2026/06/2021-p07416_1-970x647.jpg

Visual credit: gridguard_research_feeder.png

Caption: Existing workshop figure: synthetic SMART-DS Greensboro feeder udt 8894. This is not a live utility circuit.

Credit: Existing GridGuard workshop; SMART-DS synthetic Greensboro feeder udt8894

Used on: 78. GridGuard: what should this prototype prove?

Asset type: Existing scientific model plot

Transformation: Copied without alteration

Provenance note: Local existing workshop figure: Agentic_Grid_LaTeX_Source/figures/feeder.png

Evidence limit: Synthetic research feeder, not a live utility circuit or measured deployment outcome.

License or permission status: Existing user-workspace teaching figure. Source and model identity preserved.

Original source / page: research/Follow_on_workshop_source.txt

Original diagram: A generative-model call produces content

Credit: Original editable TikZ teaching diagram; source and assumptions in caption

Used on: 3. Generative AI creates content

Caption: Original example of a single generative-model call: a prompt and selected records produce a draft report for review. This illustrates one use of generative AI, not a universal boundary around the technology; the same kind of model can also participate in an agent.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/va_generation_call.svg

Editable TikZ source: diagrams/va_generation_call.tex

Concept source [DEF03]: https://www.anthropic.com/engineering/building-effective-agents

Original diagram: A valid number can still violate the tool contract

Credit: Original worked teaching illustration

Used on: 23. Unit checks catch a kV versus p.u. mistake

Caption: Illustrative typed contract. A solver expecting per-unit voltage and topology v43 rejects 12.47 kV and v42. A request with voltage 1.00 pu on base 12.47 kV and topology v43 can pass those checks. Unit conversion requires an explicit, matching voltage base and convention. Topology consistency is a separate requirement.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/typed_contract.svg

Editable TikZ source: diagrams/typed_contract.tex

Original diagram: A topology event invalidates the old study context

Credit: Original worked teaching illustration

Used on: 24. A study must match the current grid connections

Caption: Illustrative two-feeder topology, not an electrical solution. Switch S is closed in snapshot v42 at 14:35, opens at 15:02, and produces state v43. The 14:40 study used v42. At 15:05, the operator must revalidate its applicability to v43 before relying on it.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/snapshot_consistency.svg

Editable TikZ source: diagrams/snapshot_consistency.tex

Original diagram: Exercise 1 answer: C passes the first checks

Credit: Original editable TikZ worked answer for this tutorial

Used on: 27. Exercise 1 answer: C passes the first checks

Caption: The 60-second limit and packet labels belong to this teaching exercise.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/exercise_packet_solution.svg

Editable TikZ source: diagrams/exercise_packet_solution.tex

Original diagram: Three graphs with different meanings

Credit: Original editable TikZ teaching diagram; source and assumptions in caption

Used on: 34. Three kinds of graph have different jobs

Caption: Original schematics: undirected physical equipment connections, directed calculation prerequisites with fork/join dependencies, and runtime execution routes with a retry cycle. The physical network and all labels are illustrative; the diagrams are not a minimal DAG or a measured paper trace.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/va_three_graphs.svg

Editable TikZ source: diagrams/va_three_graphs.tex

Concept source [LG01]: https://docs.langchain.com/oss/python/langgraph/graph-api

Concept source [D01]: https://www.manuscriptlink.com/society/kics/media?key=kics%2Fconference%2Ficaiic2025%2Fpresentation%2F1571098105.pdf

Original diagram: A fault-record DAG follows the data prerequisites

Credit: Original editable TikZ teaching diagram; see caption for source and assumptions

Used on: 35. A DAG shows which result must come first

Caption: Original teaching adaptation of DrAgent's domain tools, not its measured execution trace. A frozen COMTRADE record R17 feeds independent find_fault_inception and find_max_fault_current computations (abbreviated in the drawing). The inception sample and sampling rate feed a time conversion; the report waits for both numerical results.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/power_fault_dag.svg

Editable TikZ source: diagrams/power_fault_dag.tex

Concept source [D01]: https://www.manuscriptlink.com/society/kics/media?key=kics%2Fconference%2Ficaiic2025%2Fpresentation%2F1571098105.pdf

Original diagram: Check the prerequisite before executing a state-dependent tool

Credit: Original native TikZ educational diagram; source and assumptions in caption

Used on: 41. PowerDAG: solve the grid before checking its voltage

Caption: Original teaching adaptation of PowerDAG v4, Section IV-C, PDF p.5. A planner proposes a voltage-violation check. A prerequisite gate blocks execution and returns the missing solve to the planner when the required solved network state is unavailable; otherwise the check proceeds. This is neither exact source code nor the complete/minimal workflow DAG.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/revision_powerdag_gate.svg

Editable TikZ source: diagrams/revision_powerdag_gate.tex

Concept source [PD01]: https://arxiv.org/pdf/2603.17418v4

Original diagram: A changed input determines the stages that rerun

Credit: Original editable TikZ teaching diagram; see caption for source and assumptions

Used on: 52. GridMind: which changes need a new run?

Caption: Original affected-stage diagram derived from GridMind 2026 Table 1, PDF p.4. Changing cascade fan-out k from 50 to 200 reruns cascade only. Shifting forecast initialization by six hours reruns all five stages: line-wind, outages, ACOPF, contingency and cascade. These are reported dependency effects, not a newly executed workflow.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/power_gridmind_invalidation.svg

Editable TikZ source: diagrams/power_gridmind_invalidation.tex

Concept source [P03]: https://openreview.net/forum?id=Cs5TBzzNym

Original diagram: Exercise 2 answer: rerun what changed

Credit: Original editable TikZ worked answer for this tutorial

Used on: 55. Exercise 2 answer: rerun what changed

Caption: Reuse follows the cited workflow's input dependencies; another implementation can differ.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/exercise_reuse_solution.svg

Editable TikZ source: diagrams/exercise_reuse_solution.tex

Concept source [P03]: https://openreview.net/forum?id=Cs5TBzzNym

Concept source [LG02]: https://docs.langchain.com/oss/python/langgraph/persistence

Original diagram: Retrieved instructions cannot acquire command authority

Credit: Original worked teaching illustration

Used on: 74. A retrieved document cannot give itself permission

Caption: Illustrative prompt-injection mechanism. A retrieved document contains an instruction to ignore limits and execute. Its contents remain data for retrieval and proposal formation. An independent policy and permission gate stands before the command service. The direct path from document text to command execution is blocked.

Format: Native TikZ in the LaTeX edition; vector SVG in the web edition

SVG diagram: diagrams/trust_boundary.svg

Editable TikZ source: diagrams/trust_boundary.tex

Concept source [T09]: https://modelcontextprotocol.io/docs/2026-07-28/tutorials/security/security_best_practices